



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMBINED SCIENCE

4003/1

PAPER 1 Multiple Choice

JUNE 2019 SESSION

1 hour

Additional materials:

- Multiple Choice answer sheet
- Soft clean eraser
- Soft pencil (type B or HB is recommended.)
- Calculator (Optional)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C and D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

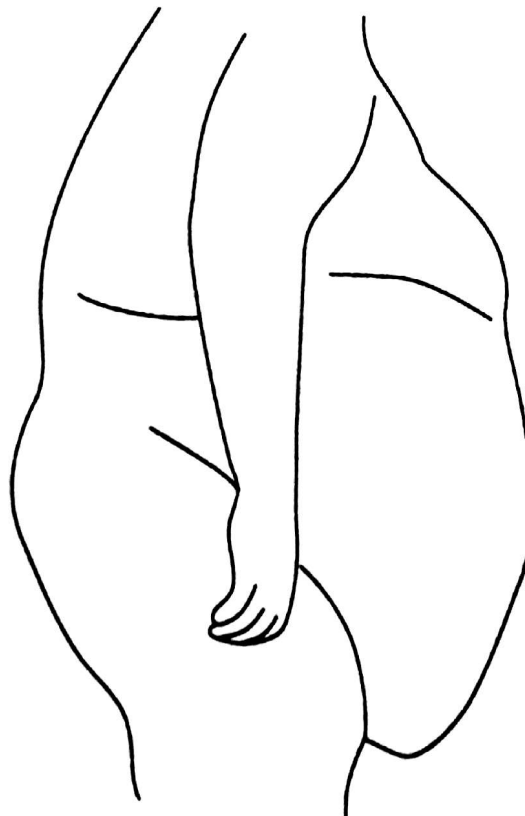
This question paper consists of 20 printed pages.

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1. Which nutrient provides energy?
 - A carbohydrate
 - B vitamin C
 - C fibre
 - D iron

2. What is the function of the gall bladder?
 - A stores bile
 - B stores urine
 - C produces bile
 - D produces urea

3. The photograph shows a condition due to malnutrition.



What is the name of the condition?

- A obesity
- B diabetes
- C kwashiorkor
- D anorexia nervosa

4. Benedict's solution was added to a food sample. The mixture was heated. A brick-red colour was observed.

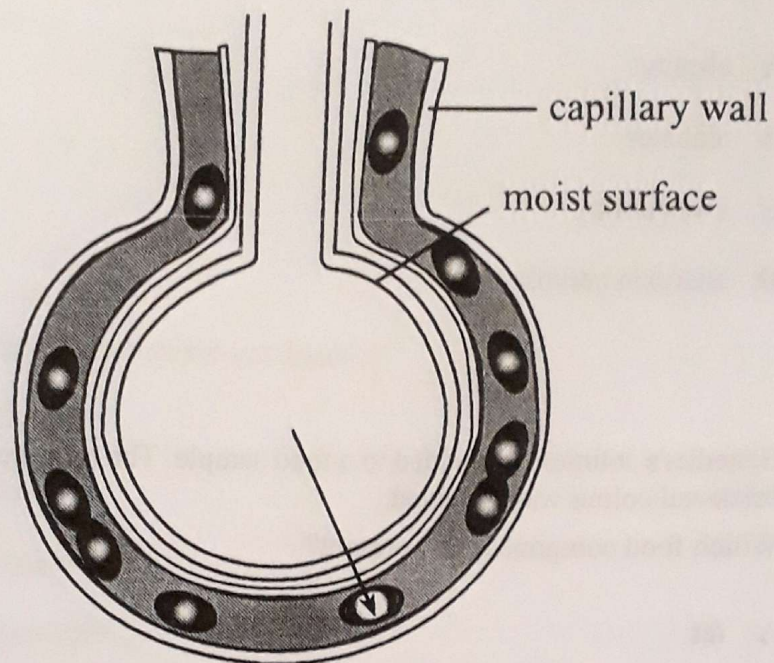
Which food component was present?

- A fat
- B starch
- C protein
- D glucose

5. During anaerobic respiration,

- A oxygen is used.
- B alcohol is produced.
- C lactic acid is produced in plant cells.
- D a large amount of energy is released.

6. The diagram shows the structure of an alveolus.

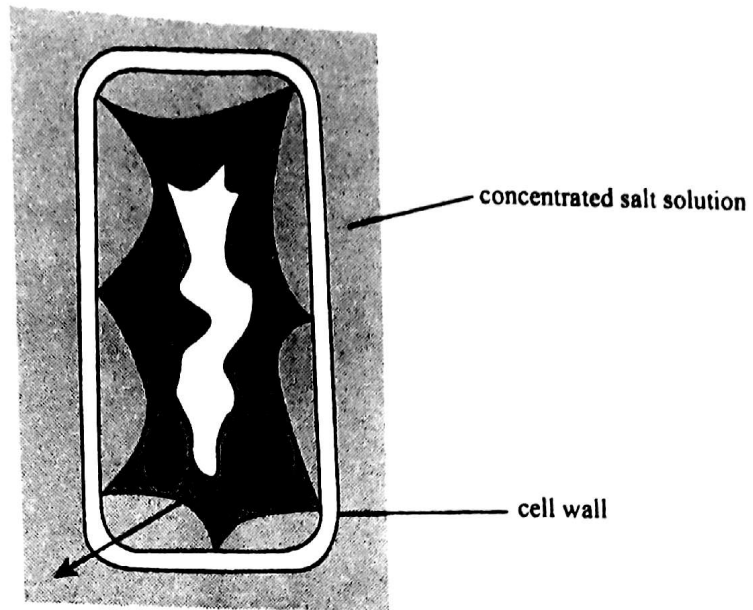


Which gas moves in the direction of the arrow?

- A carbon monoxide
- B carbon dioxide
- C nitrogen
- D oxygen

7.

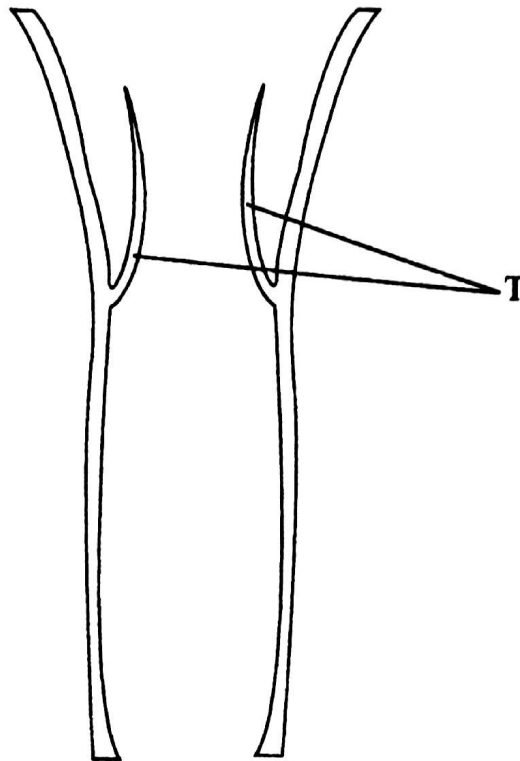
The diagram shows a plant cell after it has been placed in a concentrated salt solution.



Which substance moves in the direction of the arrow?

- A salt
- B ions
- C water
- D cytoplasm

8. The diagram shows the internal section of a blood vessel.

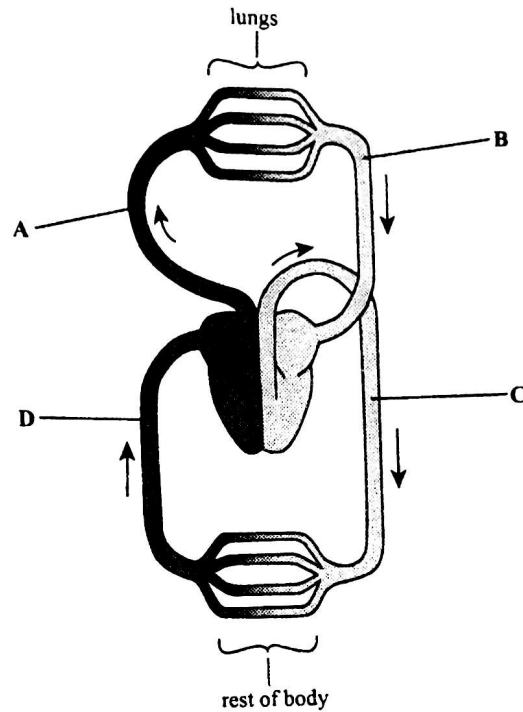


What is the function of T?

- A** to increase blood flow towards the heart
- B** to increase the lumen of the blood vessel
- C** to push blood towards the heart
- D** to prevent back flow of blood

9.

The diagram shows the human circulatory system.
Which blood vessel, **A**, **B**, **C** or **D**, has blood under highest pressure?



10.

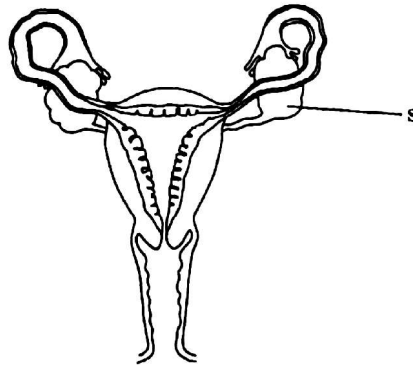
A woman starts her menstrual flow on the 2nd of April.
When is she expected to ovulate?

- A** 7 April
- B** 12 April
- C** 16 April
- D** 2 May

11. Which part of the male reproductive system stores sperms?

- A prostate gland
- B sperm duct
- C epididymis
- D testis

12. The diagram shows the reproductive system of a woman.



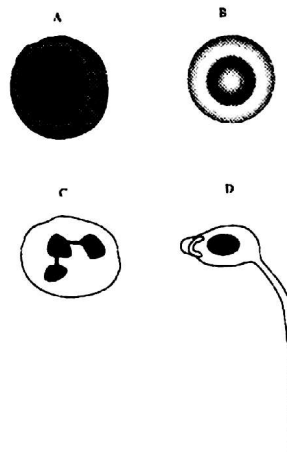
What is part S?

- A cervix
- B ovary
- C uterus
- D oviduct

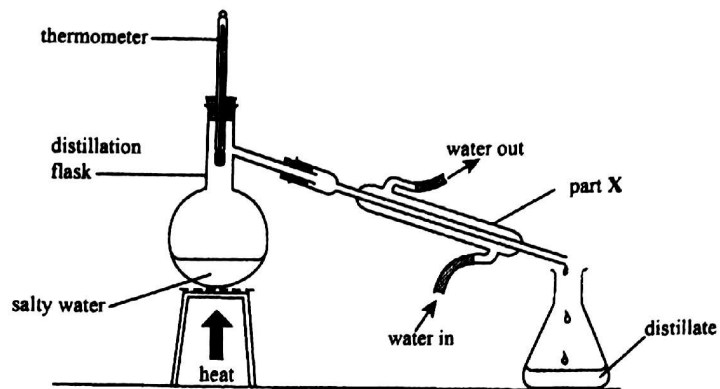
13. Which disease is spread by drinking contaminated water?

- A ebola
- B typhoid
- C malaria
- D chancroid

14. The diagram shows some specialised human cells.
Which cell, A, B, C or D, is the target of HIV?



15. The diagram shows a simple distillation apparatus.



What happens in part X?

- A the distillate is warmed
 - B vapour is converted to liquid
 - C vapour is directed into the container
 - D salt and the distillate are separated
16. Element X has an electronic configuration of 2, 8, 2.
What is the charge of an ion of X?

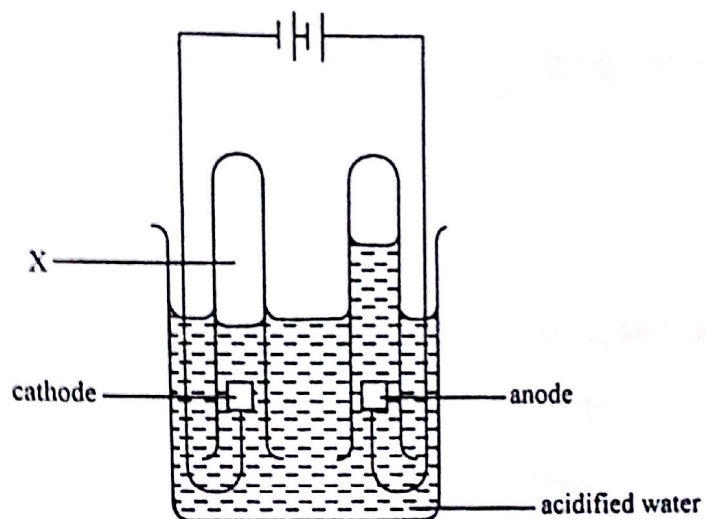
- A X^{2+}
- B X^{2-}
- C X^{6+}
- D X^{6-}

17. Which statement is true about the atom ${}_{17}^{37}\text{X}$?
- A It has 37 neutrons.
 - B It has 37 protons.
 - C It has 20 protons.
 - D It has 20 neutrons.
18. Which formula is used to calculate the concentration of a solution?
- A number of moles X volume
 - B mass X volume
 - C $\frac{\text{number of moles}}{\text{volume}}$
 - D $\frac{\text{volume}}{\text{mass}}$
19. Which one is a property of a molten ionic compound?
- A It is an electrolyte.
 - B It is insoluble in water.
 - C It has a low melting point.
 - D It is a non conductor of electricity.

20. Chlorine gas is used in
- A food preservation.
 - B electric light bulbs.
 - C soap making.
 - D water purification.
21. The reaction between sodium hydroxide and hydrochloric acid produces a salt and
- A water.
 - B a base.
 - C an acid.
 - D hydrogen.
22. Iron, copper, zinc and magnesium are all metals.
Which one is the least reactive?
- A magnesium
 - B copper
 - C zinc
 - D iron

23.

The diagram shows the electrolysis of water.



What is gas X?

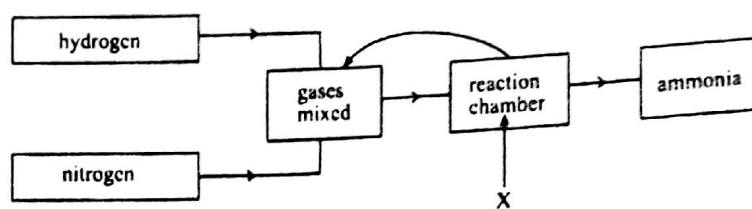
- A oxygen
- B chlorine
- C nitrogen
- D hydrogen

24.

Which process is prevented by galvanising?

- A decomposition
- B neutralisation
- C reduction
- D rusting

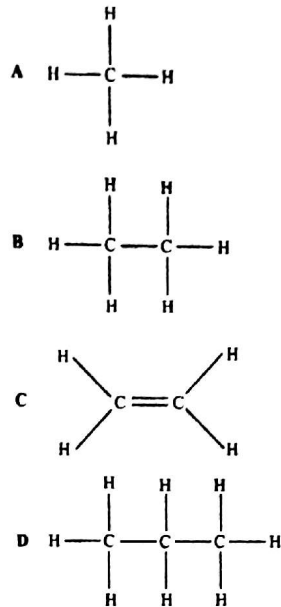
25. The diagram shows stages in the manufacture of ammonia.



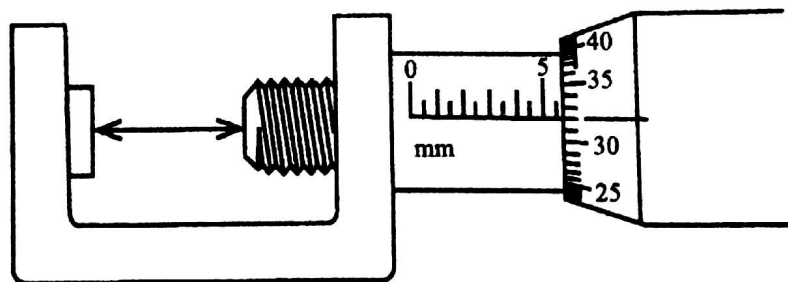
What is the pressure required at X?

- A 8 – 10 atm
 - B 100 – 150 atm
 - C 200 – 300 atm
 - D 450 – 500 atm
26. Iron is extracted from its ore in the blast furnace by the process of
- A oxidation.
 - B reduction.
 - C electrolysis.
 - D neutralisation.

27. Which one is the correct structural formula of ethane?

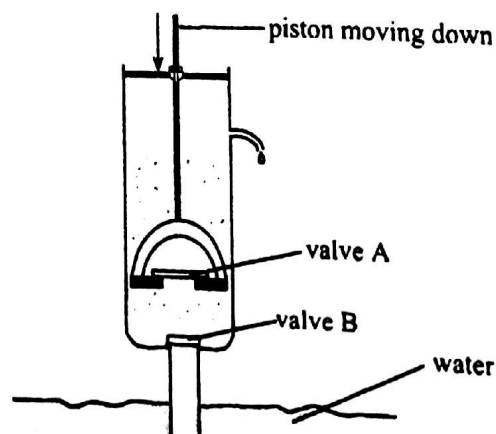


28. What is the reading shown by the micrometer screw gauge?



- A 5.50 mm
B 5.32 mm
C 5.82 mm
D 6.32 mm

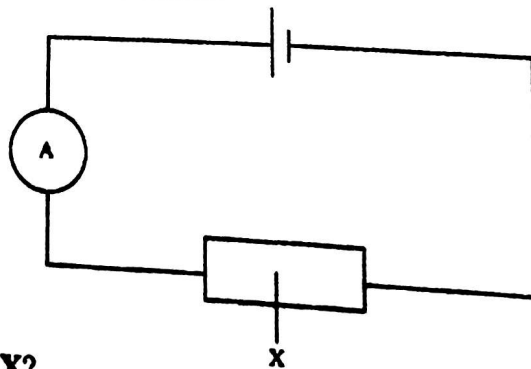
29. What is the unit of force?
- A watt
 - B joule
 - C ampere
 - D newton
30. A load of 900 N is raised 1 m by an effort of 300 N along an inclined plane. The inclined plane is 4 m long.
What is the efficiency of the inclined plane?
- A 25%
 - B 33%
 - C 67%
 - D 75%
31. The diagram shows a lift pump.



What happens to the valves during the downward stroke?

- A valve A opens, valve B closes
 B valve B opens, valve A closes
 C valves A and B open
 D valves A and B close
32. Solids transfer heat by
- A convection.
 B absorption.
 C conduction.
 D radiation.
33. The tubes inside solar heating panels use the sun's heat energy to warm water. Why are the tubes painted black?
- A The black colour is a bad emitter of heat.
 B The black colour is a bad conductor of heat.
 C The black colour is a good absorber of heat.
 D The black colour is a good reflector of heat.
34. Which energy conversion takes place in a thermal power generator?
- A chemical $\longrightarrow$ kinetic $\longrightarrow$ electrical
 B chemical $\longrightarrow$ heat $\longrightarrow$ kinetic $\longrightarrow$ electrical
 C gravitational potential $\longrightarrow$ kinetic $\longrightarrow$ electrical
 D gravitational potential $\longrightarrow$ heat $\longrightarrow$ kinetic $\longrightarrow$ electrical

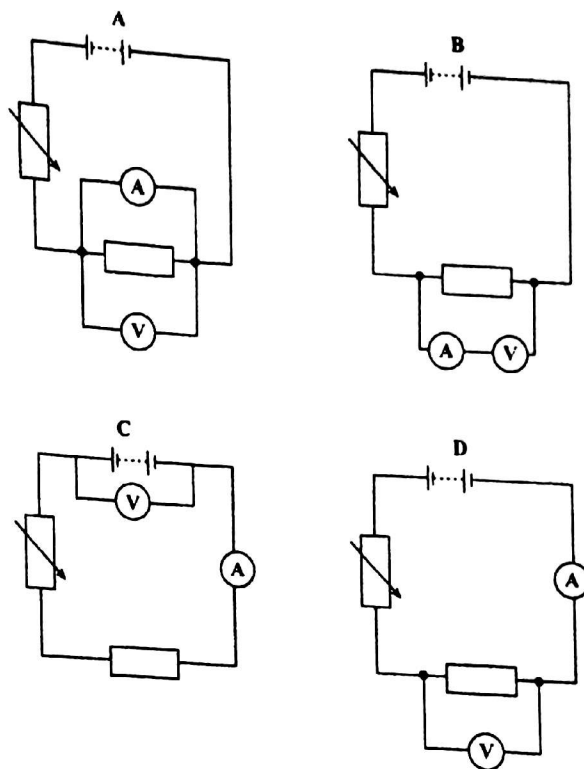
35. The speed of an electric motor can be increased by
- A using a thinner wire.
 - B reducing the size of current.
 - C increasing the number of turns on the coil.
 - D reversing the direction of the magnetic field.
36. Messages are sent through cellphones in the form of
- A heat waves.
 - B longitudinal waves.
 - C electrostatic waves.
 - D electromagnetic waves.
37. The diagram shows an electric circuit.



What is component X?

- A fuse
- B switch
- C resistor
- D ammeter

38. Which circuit, A, B, C or D, is used to verify Ohm's law?



39. What is the power of a lamp rated 12 V, 2 A?

- A 24 W
- B 14 W
- C 10 W
- D 6 W

40. What might cause an electric shock?
- A touching electrical appliances with wet hands
 - B overheating of cables for various reasons
 - C using thick electrical wires
 - D connecting an earth wire